

1 Study Cubicle Booking RDBS

Name your Jupyter Notebook as

Task_1_<class>_<index number>_<name>.ipynb

To support J2 students in their studies for A Levels, River Valley Hypothetical School is setting up study cubicles for students to book for night study. Students will need to register before being allowed to book. Each cubicle booking is for a night on a selected date for a registered student.

To help manage the booking process, RdeV is engaged to help program the booking system. RdeV decided to use a relational database to store records of students' details, cubicle details, and booking details. The database will have 3 tables to store data on (1) **student**; (2) **cubicle**; and (3) **booking**. All fields cannot be left empty.

student:

- `id` – unique student identification. for example, `rv0001`.
- `name` – name of student.
- `contact` – contact number of student. Not unique as siblings may share contact numbers.

cubicle:

- `cubicle_no` – unique cubicle identification. for example, `c01`.
- `maintenance` – status to indicate if a cubicle is under maintenance.
 - 1 indicates that cubicle is under maintenance.
 - 0 indicates that cubicle is not under maintenance.database will need to check for validity of entry.
defaults to 0 on new cubicle entry.

booking:

- `date` – date for which a cubicle is booked for. In *DDMMYYYY* format.
for example, `18112024`.
database will not be checking for validity of entry.
- `cubicle_no` – identification of cubicle that was booked.
- `student_id` – identification of student who made the booking.

[RVHS/2024/Q1 Cont']

For each of the task 1.1 to 1.4, complete the codes in Jupyter Notebook and add a comment statement at the beginning of the code using the hash symbol #, to indicate the sub-task the program code belongs to, for example

In [1]:

```
#Task 1.1  
Program code
```

Output:

Task 1.1

Write a Python program that uses SQL code to create database `cu_booking.db` with the 3 tables given. The database is in the same directory as this Jupyter Notebook. Define the primary and foreign keys for each table in your codes. [8]

Task 1.2

A sample dataset was generated for the different tables for testing the database and the programs using it. The csv files `student.csv`, `cubicle.csv` and `booking.csv` store the comma-separated values for each of the tables in the database corresponding to the table names.

Write a Python program that uses SQL code to read in the data from each file and then store each item of the data in the correct place in the database. [5]

Task 1.3

The status of maintenance needs to be updated if a cubicle is damaged or is repaired.

Write a Python program that uses SQL code to update the `maintenance status` of `c01` to 0. [2]

Task 1.4

Part of the system is to give feedback to administrators the availability of the cubicles on night of a specific date. A cubicle is available on the date only if it is not under maintenance and not book

Write a Python program that uses SQL code to print out a list of `cubicle_no` that is available on the date `'17112024'`. [4]

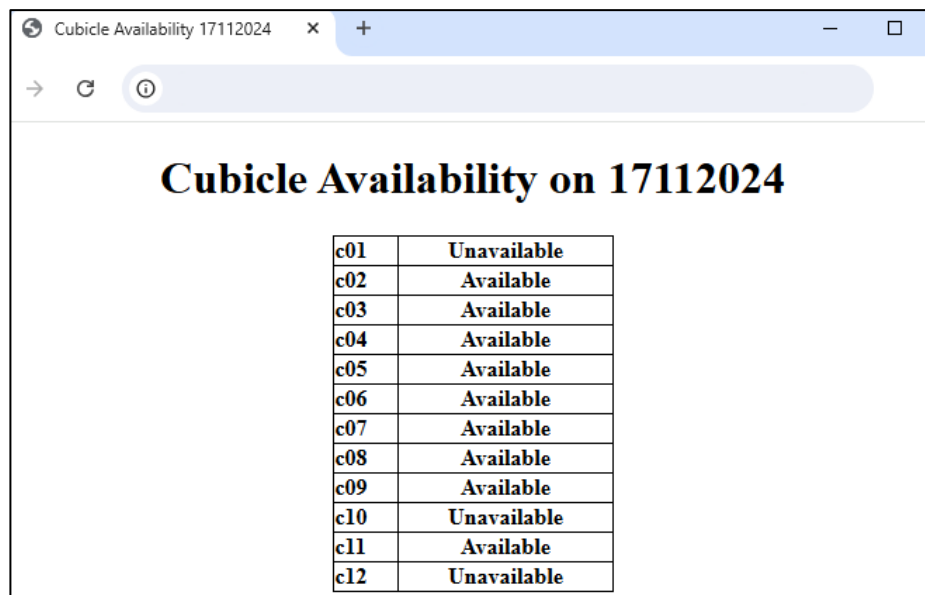
Task 1.5

The system will be implemented as a web application utilising python flask microframework.

You are tasked to write part of the web application to display the availability of cubicles on a given date. Your part of the web application will be on an URL sub-path `/available/` from its root path. The URL sub-path will have a `date` parameter to display information on cubicle availability for that date.

Eg `.../available/17112024` will display information on cubicle availability on 17112024.

Your web application will return a HTML page to display the header and the table with the associated cubicle availability information with the URL sub-path and the URL parameter as shown:



The screenshot shows a web browser window with the title 'Cubicle Availability 17112024'. The page content includes a heading 'Cubicle Availability on 17112024' and a table with 12 rows. The table has two columns: the first column contains cubicle IDs (c01 to c12) and the second column contains their availability status (Unavailable or Available). The text in the table is bold and centered.

Cubicle Availability on 17112024	
c01	Unavailable
c02	Available
c03	Available
c04	Available
c05	Available
c06	Available
c07	Available
c08	Available
c09	Available
c10	Unavailable
c11	Available
c12	Unavailable

The HTML must look similar to the one shown above with the following details:

1. HTML title and webpage header (h1 size) are the same.
2. Text within the table is all bold and must be aligned as shown.
3. Table width is 200 pixels with black 1-pixel thick border and aligned to the center of the web page.

Run your web application for date 17112024.

Save your program code as

`Task_1_5_<class>_<index>_<name>.py`

With any additional files / subfolders as needed in a folder named

`Task_1_5_<class>_<index>_<name>`

Run the web application and save the returned pages as

`Task_1_5_<class>_<index>_<name>_<page name>.html`